Solutions to Exercises from *Topics in Random Matrix Theory*

Hongkai Zhu

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1 Preparatory material

1.1 A review of probability theory

1.1.1 Foundations

No exercise.

1.1.2 Random variables

Exercise 1.1.1. (Borel-Cantelli lemma). Let $E_1, E_2, \ldots$ be a sequence of events such that $\sum_i \mathbb{P}(E_i) < \infty$. Show that almost surely, at most finitely many of the events E_i occur at once. State and prove a result to the effect that the condition $\sum_i \mathbb{P}(E_i) < \infty$ cannot be weakened.

Solution. *Proof.* Notice that

$$\begin{aligned}\{E_i \text{ i.o.}\} &= \{\text{for all } m \geq 1, \text{ there exists } n \geq m \text{ such that } E_n \text{ occurs.}\} \\ &= \left\{ \bigcap_{m=1}^{\infty} \bigcup_{n \geq m} E_n \right\} = \limsup_{n \to \infty} E_n.\end{aligned}$$

Define

$$F_m := \bigcup_{n \geq m} E_n.$$

Then

$$F_1 \supset F_2 \supset \cdots.$$

By continuity from below,

$$\mathbb{P}(E_i \text{ i.o.}) = \mathbb{P}\left(\bigcap_{m=1}^{\infty} F_m \right) = \lim_{m \to \infty} \mathbb{P}(F_m).$$

Then by union bound,

$$\mathbb{P}(F_m) = \mathbb{P}\left(\bigcup_{n \geq m} E_n \right) \leq \sum_{n \geq m} \mathbb{P}(E_n) \to 0.$$

□

to be continued.

Exercise 1.1.2. Show that these properties genuinely are in decreasing order of strength, i.e., that each property on the list implies the next.

(i) X is *surely bounded* if there exists an $M > 0$ such that $|X| \leq M$ surely.

(ii) X is *almost surely bounded* if there exists an $M > 0$ such that $|X| \leq M$ almost surely.

(iii) X is *sub-Gaussian* if there exists $C, c > 0$ such that $\mathbb{P}(|X| \geq \lambda) \leq C \exp(-c\lambda^2)$ for all $\lambda > 0$.

(iv) X has *sub-exponential tail* if there exist $C, c, a > 0$ such that $\mathbb{P}(|X| \geq \lambda) \leq C \exp(-c\lambda^a)$ for all $\lambda > 0$.

(v) X has *finite* k-th *moment* for some $k \geq 1$ if there exists C csuch that $\mathbb{E}|X|^k \leq C$.

(vi) X is *absolutely integrable* if $\mathbb{E}|X| < \infty$.

(vii) X is *almost surely finite* if $|X| < \infty$ almost surely.

Solution. *Proof.* (i) $\Longrightarrow$ (ii): trivial.

(ii) $\Longrightarrow$ (iii): let $|X| \leq M$ a.s. for some $M > 0$. Then for any $\lambda > 0$, if $\lambda > M$, $\mathbb{P}(|X| \geq \lambda) = 0$; if $0 < \lambda \leq M$, then $\mathbb{P}(|X| \geq \lambda) \leq 1$. We only need to find constant $C, c > 0$ such that

$$1 \leq C \exp(-c\lambda^2) \quad \text{for all } 0 < \lambda \leq M.$$

Since

$$\exp(-c\lambda^2) \geq \exp(-cM^2)$$

on $[0, M]$, it suffices to choose $1 \leq C \exp(-cM^2)$, which on requires

$$C \geq \exp(cM^2) < \infty.$$

(iii) $\Longrightarrow$ (iv): take $a = 2$.

(iv) $\Longrightarrow$ (v): assume there exists $C, c, a > 0$ such that $\mathbb{R}(|X| \geq \lambda) \leq C \exp(-c\lambda^a)$ for all $\lambda > 0$. Then

$$\mathbb{E}|X|^k = \int_0^\infty \mathbb{P}(|X|^k \geq \lambda)\ d\lambda = \int_0^\infty \mathbb{P}(|X| \geq \lambda^{1/k})\ d\lambda \leq C \int_0^\infty \exp(-c\lambda^{a/k})\ d\lambda.$$

Let $t = c\lambda^{a/k}$, then

$$d\lambda = \frac{k}{a} c^{-k/a} t^{k/a-1}\ dt.$$

So

$$\mathbb{E}|X|^k \leq \frac{Ck}{a} c^{-k/a} \int_0^\infty t^{k/a-1} e^{-t}\ dt = \frac{Ck}{a} c^{-k/a} \Gamma(k/a) < \infty.$$

(v) $\Longrightarrow$ (vi): assume there exists C such that $\mathbb{E}|X|^k \leq C$ for some k. Then

$$\mathbb{E}|X| = \int_0^\infty \mathbb{P}(|X| \geq \lambda)\ d\lambda = \int_0^\infty \mathbb{P}(|X|^k \geq \lambda^k)\ d\lambda.$$

By the Markov's inequality,

$$\mathbb{E}|X| \leq \int_0^\infty \lambda^{-k} \mathbb{E}|X|^k\ d\lambda = \mathbb{E}|X|^k \int_0^\infty \lambda^{-k}\ d\lambda = \mathbb{E}|X|^k \frac{1}{k-1} < \infty.$$

Alternatively, by the Hölder's inequality with $p = k$ and $q = \frac{k}{k-1}$,

$$\mathbb{E}|X| = (\mathbb{E}|X|^k)^{1/k} (\mathbb{E}1^q)^{1/q} = (\mathbb{E}|X|^k)^{1/k} < \infty.$$

Alternatively, using the inequality

$$x \leq 1 + x^k,$$

which can be seen from: if $0 \leq x \leq 1$,, then $x \leq 1$; if $x \geq 1$, then $x \leq x^k$. Let $x = X$ and taking expectations, we have

$$\mathbb{E}|X| \leq 1 + \mathbb{E}|X|^k < \infty.$$

(vi) $\Longrightarrow$ (vii): Assume $\mathbb{E}|X| < \infty$, then

$$\mathbb{P}(|X| < \infty) = \lim_{\lambda \to \infty} \mathbb{P}(|X| < \lambda) = \lim_{\lambda \to \infty} (1 - \mathbb{P}(|X| < \lambda)) \geq \lim_{\lambda \to \infty} \left(1 - \frac{\mathbb{E}|X|}{\lambda}\right) = 1$$

by the Markov's inequality. So

$$\mathbb{P}(|X| < \infty) = 1.$$

□

Exercise 1.1.3. Show that each of these properties are closed under vector space operations.

Solution. *Proof.* (i) Suppose $|X| \le M_X$ and $|Y| \le M_Y$, then

$$|cX| = |c||X| \le |c|M_X$$

And

$$|X+Y| \le |X| + |Y| \le M_X + M_Y.$$

(ii) Suppose $\mathbb{P}(|X| \le M_X) = \mathbb{P}(|Y| \le M_Y) = 1$. Let $M = \max(M_X, M_Y)$, then

$$\mathbb{P}(|cX| \le |c|M_X) = \mathbb{P}(|X| \le M_X) = 1.$$

And

$$\mathbb{P}(|X+Y| \le 2M) \ge \mathbb{P}(|X|+|Y| \le 2|M|) = \mathbb{P}(|X| \le M \cup |Y| \le M) \ge \mathbb{P}(|X| \le M) \ge \mathbb{P}(|X| \le M_X) = 1$$

so

$$\mathbb{P}(|X+Y| \le 2M) = 1.$$

(iii) Suppose there exists $C, c > 0$ such that $\mathbb{P}(|X| \ge \lambda) \le C\exp(-c\lambda^2)$ for all $\lambda > 0$, then

$$\mathbb{P}(|c'X| \ge c'\lambda) \le C\exp(-cc'^2\lambda^2)$$

for all $\lambda > 0$. And

$$\begin{aligned}
\mathbb{P}(|X+Y| \ge \lambda) &\le \mathbb{P}(|X| + |Y| \ge \lambda) \\
&= \mathbb{P}(|X| \ge \lambda/2 \cup |Y| \ge \lambda/2) \\
&\le \mathbb{P}(|X| \ge \lambda/2) + \mathbb{P}(|Y| \ge \lambda/2) \\
&\le C_X \exp(-c_X\lambda^2/4) + C_Y \exp(-c_Y\lambda^2/4) \le (C_X + C_Y)\exp(-c\lambda^2)
\end{aligned}$$

if we take $c = \min(c_X, c_Y)/4$.

(iv) Similar with (iii).

(v) Suppose $\mathbb{E}|X|^k \le C_X$ and $\mathbb{E}|Y|^l \le C_Y$. Then

$$\mathbb{E}|cX|^k = |c|^k\mathbb{E}|X|^k \le c^k C_X.$$

Let $p = \min(k, l)$, we have

$$\mathbb{E}|X+Y|^p \le \mathbb{E}|X|^p + \mathbb{E}|Y|^p \le \mathbb{E}|X|^k + \mathbb{E}|Y|^l \le C_X + C_Y.$$

(vi) Suppose $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$. Then $\mathbb{E}|cX| = |c|\mathbb{E}|X| < \infty$ and $\mathbb{E}|X+Y| \le \mathbb{E}|X| + \mathbb{E}|Y| < \infty$.

(vii) Suppose $\mathbb{P}(|X| < \infty) = \mathbb{P}(|Y| < \infty) = 1$. Then $\mathbb{P}(|cX| < \infty) = \mathbb{P}(|X| < \infty) = 1$. And

$$\mathbb{P}(|X+Y| < \infty) = \mathbb{P}(|X| < \infty \cap |Y| < \infty) \ge 1 - \mathbb{P}(|X| = \infty) - \mathbb{P}(|Y| = \infty) = 1$$

so

$$\mathbb{P}(|X+Y| < \infty) = 1.$$

□

Exercise 1.1.4. Show that a real-valued random variable X is sub-Gaussian if and only if there exists $C > 0$ such that $\mathbb{E}e^{tX} \le C\exp(Ct^2)$ for all real t, and if and only if there exists $C > 0$ such that $\mathbb{E}|X|^k \le (Ck)^{k/2}$ for all $k \ge 1$.

Exercise 1.1.5. Show that a real-valued random variable X has sub-exponential tails if and only if there exists $C > 0$ such that $\mathbb{E}|X|^k \le \exp(Ck^C)$ for all positive integers k.

Solution. *Proof.* ($\Rightarrow$) Suppose X has sub-exponential tails, i.e. exists C, c, a such that

$$\mathbb{P}(|X| \geq \lambda) \leq C \exp(-c\lambda^a)$$

for all $\lambda > 0$. Then

$$\mathbb{E}|X|^k = \int_0^\infty \mathbb{P}(|X|^k \geq \lambda)\ d\lambda = \int_0^\infty \mathbb{P}(|X| \geq \lambda^{1/k}\ d\lambda \leq \int_0^\infty C \exp(-c\lambda^{a/k})\ d\lambda.$$

Let $u = c\lambda^{a/k}$, then $\lambda = (u/c)^{k/a}$ and $d\lambda = \frac{k}{a} c^{-k/a} u^{k/a-1}\ du$. Thus

$$\mathbb{E}|X|^k = \frac{Ckc^{-k/a}}{a} \int_0^\infty u^{k/a-1} e^{-u}\ du = \frac{Ckc^{-k/a}}{a} \Gamma(k/a).$$

□

to be continued

Exercise 1.1.7. We say that a real number $\mathbb{M}(X)$ is a *median* of a real-valued random variable X if $\mathbb{P}(X > \mathbb{M}(X)), \mathbb{P}(X < \mathbb{M}(X)) \leq 1/2$.

(i) Show that a median always exists, and if X is absolutely continuous with strictly positive density function, then the median is unique.

(ii) If X has finite second moment, show that

$$\mathbb{M}(X) = \mathbb{E}(X) + O(\mathrm{Var}(X)^{1/2})$$

for any median $\mathbb{M}(X)$.

Solution. *Proof.* (i) WLOG suppose for any x we have $\mathbb{P}(X > x) > 1/2$ and $\mathbb{P}(X < x) \leq 1/2$. Then

$$\mathbb{P}(X < x) \leq \mathbb{P}(X \leq x) = 1 - \mathbb{P}(X > x) \leq 1/2$$

which contradict to the assumption. If there exists $M = \mathbb{M}(X)$ and $N = \mathbb{N}(X)$ both being medians of X, then $\mathbb{P}(X > N), \mathbb{P}(X < M) \leq 1/2$. WLOG assume $N < M$, then

$$\mathbb{P}(X \leq N) = 1 - \mathbb{P}(X > N) > 1/2 \quad \text{and} \quad \mathbb{P}(X \geq M) = 1 - \mathbb{P}(X < M) > 1/2.$$

Thus

$$\mathbb{P}(N < X < M) = 1 - \mathbb{P}(X \leq N) - \mathbb{P}(X \geq M) < 1 - 1/2 - 1/2 = 0$$

which is impossible, so $N = M$.

□

to be continued

Exercise 1.1.8. (Jensen's inequality). Let $F : \mathbb{R} \to \mathbb{R}$ be a convex function (thus $F((1-t)x+ty) \leq (1-t)F(x)+tF(y)$ for all $x, y \in \mathbb{R}$ and $0 \leq t \leq 1$), and let X be a bounded real-valued random variable. Show that $\mathbb{E}F(X) \geq F(\mathbb{E}X)$. (*Hint:* Bound F from below using a tangent line at $\mathbb{E}X$.) Extend this inequality to the case when X takes values in $\mathbb{R}^n$ (and F and $\mathbb{R}^n$ as its domain.)

Solution. *Proof.* For every $a \in R$, a convex F is bounded below by its tangent line at a due to convexity, i.e.

$$F(x) \geq F'(a)(x-a) + F(a)$$

for all $x, a \in R$. Let $a = \mathbb{E}X$ and taking expectations, we have

$$\begin{aligned}\mathbb{E}(F(X)) &\geq \mathbb{E}(F'(\mathbb{E}X)(X - \mathbb{E}X) + F(\mathbb{E}X)) \\ &= F'(\mathbb{E}X)\mathbb{E}X - F'(\mathbb{E}X)\mathbb{E}X + F(\mathbb{E}X) = F(\mathbb{E}X).\end{aligned}$$

□

Extension to $\mathbb{R}^n$: Let $F : \mathbb{R}^n \to \mathbb{R}$ be a convex function (thus $F((1-t)x+ty) \leq (1-t)F(x)+tF(y)$ for all $x, y \in \mathbb{R}$ and $0 \leq t \leq 1$), and let X be a bounded real-valued random variable. Then $\mathbb{E}F(X) \geq F(\mathbb{E}X)$.

if F is not differentiable, how do we get F'?

Exercise 1.1.9. (Paley-Zygmund inequality). Let X be a positive random variable with finite variance. Show that

$$\mathbb{P}(X \geq \lambda \mathbb{E}(X)) \geq (1-\lambda)^2 \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}$$

for any $0 < \lambda < 1$.

Solution. *Proof.* We split $\mathbb{E}X$ into

$$\mathbb{E}X = \mathbb{E}X\mathbb{1}_{(X<\lambda\mathbb{E}X)} + \mathbb{E}X\mathbb{1}_{(X\geq\lambda\mathbb{E}X)}.$$

Since

$$\mathbb{E}X\mathbb{1}_{(X<\lambda\mathbb{E}X)} < \mathbb{E}(\lambda\mathbb{E}X) = \lambda\mathbb{E}X,$$

and by the Cauchy-Schwartz inequality,

$$\mathbb{E}X\mathbb{1}_{(X\geq\lambda\mathbb{E}X)} \leq (\mathbb{E}X^2\mathbb{E}\mathbb{1}^2_{(X\geq\lambda\mathbb{E}X)})^{1/2} = (\mathbb{E}X^2\mathbb{P}(X \geq \lambda\mathbb{E}X))^{1/2}$$

since $\mathbb{1}^2_A = \mathbb{1}_A$. Thus

$$\mathbb{E}X < \lambda\mathbb{E}X + (\mathbb{E}X^2\mathbb{P}(X \geq \lambda\mathbb{E}X))^{1/2}.$$

Rearranging the inequality will give the desired form. □

How can we get equality?

1.1.3 Independence

Exercise 1.1.10 Let X, X' be random variables (on sample spaces Ω, Ω' respectively) taking values in a range R, such that $X \equiv X'$. Show that after extending the spaces Ω, Ω', the two random variables X, X' are isomorphic, in the sense that there exists a probability space isomorphism $\pi : \Omega \to \Omega'$ (i.e., an invertible extension map whose inverse is also an extension map) such that $X = X' \circ \pi$.

Solution. *Let $\overline{\Omega} = \Omega \times [0,1]$ and $\overline{\Omega'} = \Omega' \times [0,1]$ and equip $[0,1]$ with Lebesgue measure. Then the random variables are extended naturally:*

$$\overline{X}(\omega, u) = X(\omega) \quad \text{and} \quad \overline{X'}(\omega', u') = X'(\omega').$$

For each $r \in R$, let $\Omega_r = X^{-1}(r)$ and $\Omega'_r = (X')^{-1}(r)$. Since $X \equiv X'$, $\mathbb{P}(\Omega_r) = \mathbb{P}'(\Omega'_r)$. We define $\pi : \overline{\Omega} \to \overline{\Omega'}$ fiberwise. For each $r \in R$, we choose a probability space isomorphism:

$$\pi_r = \overline{\Omega}_r \to \overline{\Omega'}_r.$$

It follows that

$$X'(\pi_r(\omega, u)) = r = X(\omega, u).$$

By gluing these π_r maps together across all $r \in R$, we obtain a global map π.

Exercise 1.1.11. Let $(X_\alpha)_{\alpha\in A}$ be a family of random variables, with each X_α taking values in a measurable space R_α.

(i) Show that the $(X_\alpha)_{\alpha\in A}$ are jointly independent if and only if for every collection of distinct elements $\alpha_1, \ldots, \alpha_{k'}$ of A, and all measurable subsets $E_i \subset R_{\alpha_i}$ for $1 \leq i \leq k'$, one has

$$\mathbb{P}(A_{\alpha'} \in E_i \text{ for all } 1 \leq i \leq k') = \prod_{i=1}^{k'} \mathbb{P}(X_{\alpha_i} \in E_i).$$

(ii) Show that the necessary and sufficient condition $(X_\alpha)_{\alpha\in A}$ being k-wise independent is the same, except that k' is constrained to be at most k.

Solution. *Proof.* (i) Suppose (X_α) are jointly independent. Fix distinct $\alpha_1, \ldots, \alpha_{k'}$ and measurable $E_i \subset R_{\alpha_i}$. Then the event

$$B_i := \{X_{\alpha_i} \in E_i\} = X_{\alpha_i}^{-1}(E_i)$$

lies in $\sigma(X_{\alpha_i})$. By independence of the σ-algebras,

$$\mathbb{P}(X_{\alpha_i} \in E_i \text{ for all } 1 \le i \le k') = \mathbb{P}\left(\bigcap_{i=1}^{k'} B_i\right) = \prod_{i=1}^{k'} \mathbb{P}(B_i) = \prod_{i=1}^{k'} \mathbb{P}(X_{\alpha_i} \in E_i).$$

(ii) Suppose for any distinct $\alpha_1, \ldots, \alpha_{k'}$ and measurable $E_i \subset R_{\alpha_i}$,

$$\mathbb{P}(A_{\alpha'} \in E_i \text{ for all } 1 \le i \le k') = \prod_{i=1}^{k'} \mathbb{P}(X_{\alpha_i} \in E_i).$$

Define the π-system

$$\mathcal{P}_i := \{X_{\alpha_i}^{-1}(E) : E \subset R_{\alpha_i} \text{ measurable}\}.$$

Then $\mathcal{P}_i$ is closed under finite intersections and $\sigma(\mathcal{P}_i) = \sigma(X_{\alpha_i}) = \mathcal{F}_{\alpha_i}$. Fix $B_1 \in \mathcal{P}_1, \ldots, B_{k'-1} \in \mathcal{P}_{k'-1}$. Define

$$\mathcal{L}_{k'} := \left\{ B \in \mathcal{F}_{\alpha_{k'}} : \mathbb{P}\left(\bigcap_{i=1}^{k'-1} B_i \cap B\right) = \left(\prod_{i=1}^{k'-1} \mathbb{P}(B_i)\right) \mathbb{P}(B) \right\}.$$

$\mathcal{L}_{k'}$ is a λ-system. By assumption $\mathcal{P}_{k'} \subset \mathcal{L}_{k'}$. Since $\mathcal{P}_{k'}$ is a π-system generating $\mathcal{F}_{\alpha_{k'}}$, the $\pi - \lambda$ theorem gives $\mathcal{L}_{k'} = \mathcal{F}_{\alpha_{k'}}$. Hence the desired product formula holds for all $B_{k'} \in \mathcal{F}_{\alpha_{k'}}$, still with $B_1, \ldots, B_{k'-1} \in \mathcal{P}_i$.

Fix $B_{k'} \in \mathcal{F}_{\alpha_{k'}}$ and repeat the same construction for coordinate $k' - 1$. At the end we have the product formula.

(iii) By definition, $(X_\alpha)_{\alpha \in A}$ is k-wise independent if and only if for every choice of distinct $\alpha_1, \ldots, \alpha_{k'}$ with $k' \le k$, the σ-algebras $\mathcal{F}_{\alpha_1}, \ldots, \mathcal{F}_{\alpha_{k'}}$ are independent. Applying part (i) to each finite subfamily of size $k' \le k$ gives:

$$\begin{aligned} &(X_\alpha) \text{ are } k - \text{wise independent} \\ \iff &\mathbb{P}(A_{\alpha'} \in E_i \text{ for all } 1 \le i \le k') = \prod_{i=1}^{k'} \mathbb{P}(X_{\alpha_i} \in E_i) \\ &\text{for all distinct } \alpha_1, \ldots, \alpha_{k'} \text{ with } k' \le k \text{ and measurable } E_i. \end{aligned}$$

□

Exercise 1.1.12. Let $\mathbb{F}$ be a finite field, and let V be a subspace of $\mathbb{F}^n$ for some finite n. Let $(X_1, \ldots, X_n)$ be drawn uniformly at random from V. Suppose that V is not contained in any coordinate hyperplane in $\mathbb{F}^n$.

(i) Show that each X_i, $1 \le i \le n$ is uniformly distributed in $\mathbb{F}$.

(ii) Show that for any $k \ge 2$, that $(X_1, \ldots, X_n)$ is k-wise independent if and only if V is not contained in any hyperplane which is definable using at most K of the coordinate variables.

(iii) Show that $(X_1, \ldots, X_n)$ is jointly independent if and only if $V = \mathbb{F}^n$.

1.1.4 Conditioning

Exercise 1.1.21. If E and $\overline{E}$ both occur with positive probability, establish the identities

$$\mathbb{P}(F) = \mathbb{P}(F \mid E)\mathbb{P}(E) + \mathbb{P}(F \mid \overline{E})\mathbb{P}(\overline{E})$$

for any (unconditional) event F and

$$\mu_X = \mu_{(X|E)}\mathbb{P}(E) + \mu_{(X|\overline{E})}\mathbb{P}(\overline{E})$$

for any (unconditional) random variable X (in the original sample space). In a similar spirit, if X is a non-negative or absolutely integrable scalar (unconditional) random variable, show that $(X \mid E)$, $(X \mid \overline{E})$ are also non-negative and absolutely integrable on their respective conditioned spaces, and that

$$\mathbb{E}X = \mathbb{E}(X \mid E)\mathbb{P}(E) + \mathbb{E}(X \mid \overline{E})\mathbb{P}(\overline{E}).$$

Solution. *Proof.* For any (unconditional) event F,

$$\mathbb{P}(F) = \mathbb{P}(F \cap E) + \mathbb{P}(F \cap \overline{E}) = \mathbb{P}(F \mid E)\mathbb{P}(E) + \mathbb{P}(F \mid \overline{E})\mathbb{P}(\overline{E}).$$

Then for any S,

$$\mu_X(S) = \mathbb{P}(X \in S) = \mathbb{P}(X \in S \mid E)\mathbb{P}(E) + \mathbb{P}(X \in S \mid \overline{E})\mathbb{P}(\overline{E}) = \mu_{X|E}(S)\mathbb{P}(E) + \mu_{X|\overline{E}}(S)\mathbb{P}(\overline{E}).$$

Therefore

$$\mu_X = \mu_{(X|E)}\mathbb{P}(E) + \mu_{(X|\overline{E})}\mathbb{P}(\overline{E}).$$

If X is non-negative, then clearly its restrictions to E and $\overline{E}$ are non-negative. Same applies to absolutely integrability. By

$$\mathbb{E}(X \mid E) = \frac{\mathbb{E}(X\mathbb{1}_E)}{\mathbb{P}(E)},$$

we have

$$\mathbb{E}X = \mathbb{E}X\mathbb{1}_E + \mathbb{E}\mathbb{1}_{\overline{E}} = \mathbb{E}(X \mid E)\mathbb{P}(E) + \mathbb{E}(X \mid \overline{E})\mathbb{P}(\overline{E}).$$

□

Exercise 1.1.22. Let Y be a discrete random variable with range R. Then we have

$$\mathbb{P}(F) = \sum_{y \in R} \mathbb{P}(F \mid Y = y)\mathbb{P}(Y = y)$$

for any (unconditional) event F, and

$$\mu_X = \sum_{y \in R} \mu_{(X|Y=y)}\mathbb{P}(Y = y)$$

for any (unconditional) random variable X (where the sum of non-negative measures is defined in the obvious manner), and for absolutely integrable or non-negative (unconditional) random variable X, one has

$$\mathbb{E}X = \sum_{y \in R} \mathbb{E}(X \mid Y = y)\mathbb{P}(Y = y).$$

In all of these identities, we adopt the convention that any term involving $\mathbb{P}(Y = y)$ is ignored when $\mathbb{P}(Y = y) = 0$.

Solution. *Proof.* For any event F,

$$\mathbb{P}(F) = \mathbb{P}\left(\bigcup_{y\in R} F \cap \{Y = y\}\right) = \sum_{y\in R} \mathbb{P}(F \cap \{Y = y\}) = \sum_{y\in R} \mathbb{P}(F \mid Y = y)\mathbb{P}(Y = y).$$

The second equality holds dur to disjoint union. For any measurable set S,

$$\mu_X(S) = \mathbb{P}(X \in S) = \sum_{y\in R} \mathbb{P}(X \in S \mid Y = y)\mathbb{P}(Y = y) = \sum_{y\in R} \mu_{(X|Y=y)}\mathbb{P}(Y = y).$$

Assume $X \geq 0$ and $X \in L^1$,

$$\mathbb{E}X = \mathbb{E}\left(\sum_{y\in R} X\mathbb{1}_{\{Y=y\}}\right) = \sum_{y\in R} \mathbb{E}X\mathbb{1}_{\{Y=y\}} = \sum_{y\in R} \mathbb{E}(X \mid Y = y)\mathbb{P}(Y = y).$$

□

Exercise 1.1.23. (Almost uniqueness of disintegrations). Let $(R', (\mu_y)_{y\in R'})$, $(\tilde{R}', (\tilde{\mu}_y)_{y\in \tilde{R}'})$ be two disintegrations of the same random variable Y. Show that for any event F, one has $\mathbb{P}(F \mid Y = y) = \tilde{\mathbb{P}}(F \mid Y = y)$ for μ_Y almost every $y \in R$, where the conditional probabilities $\mathbb{P}(\mid Y = y)$ and $\tilde{\mathbb{P}}(\mid Y = y)$ are defined using the disintegrations $(R', (\mu_y)_{y\in R'})$, $(\tilde{R}', (\tilde{\mu}_y)_{y\in \tilde{R}'})$, respectively. (*Hint:* Argue by contradiction, and consider the set of y for which $\mathbb{P}(F \mid Y = y)$ exceeds $\tilde{\mathbb{P}}(F \mid Y = y)$ (or vise versa0 by some fixed $\epsilon > 0$.)

Proof. Suppose

$$A = \{y : \mathbb{P}(F \mid Y = y) \geq \tilde{\mathbb{P}}(F \mid Y = y) + \epsilon\}$$

for $\epsilon > 0$. Note that

$$\mathbb{P}(F \cap \{Y \in A\}) = \int_A \mathbb{P}(F \mid Y = y)\ \mu_Y(dy) = \int_A \tilde{\mathbb{P}}(F \mid Y = y)\ \mu_Y(dy).$$

Since on A we have

$$\mathbb{P}(F \mid Y = y) \geq \tilde{\mathbb{P}}(F \mid Y = y) + \epsilon\},$$

we have

$$\int_A \mathbb{P}(F \mid Y = y)\ \mu_Y(dy) > \int_A \tilde{\mathbb{P}}(F \mid Y = y)\ \mu_Y(dy)$$

which is a contradiction. So $A = \emptyset$. □

Exercise 1.1.24 (Fubini-Tonelli theorem). Let $(R', (\mu_y)_{y\in R'})$ be a disintegration of a random variable Y taking values in a measurable space R, and let X be a non-negative (resp. absolutely integrable) scalar random variable. Show that for μ_Y-almost all $y \in R$, $(X \mid Y = y)$ is a non-negative (resp. absolutely integrable) random variable, and one has the identity

$$\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X),$$

where $\mathbb{E}(X \mid Y)$ is the (almost surely defined) random variable that equals $\mathbb{E}(X \mid Y = y)$ whenever $y \in R'$. More generally, show that

$$\mathbb{E}(\mathbb{E}(X \mid Y)f(Y)) = \mathbb{E}(Xf(X)),$$

whenever $f : R \to \mathbb{E}$ is a non-negative (resp. bounded) measurable function.

1.1.5 Convergence

To be updated...

1.2 Stirling's formula

Exercise 1.2.1 (Entropy formula). Let n be large, let $0 < \gamma < 1$ be fixed and let $1 \le m \le n$ be an integer if the form $m = (\gamma + o(1))n$. Show that $\binom{n}{m} = \exp((h(\gamma) + o(1))n)$, where $h(\gamma)$ is the *entropy function*

$$h(\gamma) := \gamma \log \frac{1}{\gamma} + (1-\gamma) \log \frac{1}{1-\gamma}.$$

Solution. *Proof.* Binomial gives

$$\binom{n}{m} = \frac{n!}{(n-m)!m!}.$$

By the Stirling's formula,

$$\binom{n}{m} = (1+o(1))\sqrt{\frac{n}{2\pi(n-m)m}} \left(\frac{n}{n-m}\right)^n \left(\frac{n-m}{m}\right)^m.$$

Since $m = (\gamma + o(1))n$,

$$A = \sqrt{\frac{n}{2\pi(n-m)m}} \sim \sqrt{\frac{1}{2\pi\gamma(1-\gamma)}} \cdot n^{-1/2},$$

so

$$A = O(n^{-1/2}).$$

For the other part, we have

$$\begin{aligned} B = \left(\frac{n}{n-m}\right)^n \left(\frac{n-m}{m}\right)^m &= \exp\left(n \log \frac{n}{n-m} + m \log \frac{n-m}{m}\right) \\ &\to \exp\left(n \log \frac{1}{1-\gamma} + \gamma n \log \frac{1-\gamma}{\gamma}\right) = \exp(h(\gamma)n). \end{aligned}$$

So

$$B = (1+o(1)) \exp(h(\gamma)n).$$

Write

$$\binom{n}{m} = (1+o(1))AB.$$

Then

$$\log \binom{n}{m} = \log A + \log B + \log(1+o(1)).$$

Since $A = O(n^{-1/2})$,

$$\log A = -\frac{1}{2} \log n + O(1).$$

So $\log A = o(n)$ since $\log n/n \to 0$. Also $\log(1+o(1)) = o(1)$ hence certainly $o(n)$, so

$$\log B = \log(1+o(1)) + h(\gamma)n = o(1) + h(\gamma)n.$$

Combining them together gives

$$\log \binom{n}{m} = o(n) + o(1) + h(\gamma)n + o(1) = h(\gamma)n + o(n).$$

Therefore

$$\binom{n}{m} = \exp(h(\gamma)n + o(n)) = \exp((h(\gamma) + o(1))n).$$

□

For m near $n/2$, one also has the following more precise bound:

Exercise 1.2.2 (Refined entropy formula). Let n be large, and let $1 \leq m \leq n$ be an integer of the form $m = n/2 + k$ for some $k = o(n^{2/3})$. Show that

$$\binom{n}{m} = \left(\sqrt{\frac{2}{\pi}} + o(1)\right) \frac{2^n}{\sqrt{n}} \exp(-2k^2/n).$$

Solution. *Proof.* Similar with above, after applying the Stirling's formula, we have

$$\binom{n}{m} = (1 + o(1))\sqrt{\frac{n}{2\pi m(n-m)}} \frac{n^n}{m^m (n-m)^{n-m}}.$$

Taking logs of the big ratio:

$$\log \frac{n^n}{m^m (n-m)^{n-m}} = n \log n - m \log m - (n-m) \log(n-m).$$

Let $p := m/n = 1/2 + \delta$ where $\delta = k/n = o(n^{-1/3})$, then this equals

$$n\left(-p \log p - (1-p)\log(1-p)\right) = nh(p).$$

So

$$\binom{n}{m} = (1 + o(1)) \frac{1}{\sqrt{2\pi np(1-p)}} \exp(nh(p)).$$

Taylor expansion at $p = 1/2$ gives

$$h(\delta + 1/2) = \log 2 - 2\delta^2 + O(\delta^4)$$

and the cubic terms vanish by symmetry $h(\delta + 1/2) = h(-\delta + 1/2)$. Multiply by n gives

$$nh(p) = n \log 2 - 2n\delta^2 + O(n\delta^4).$$

Since $n\delta^2 = k^2/n$ and $n\delta^4 = k^4/n^3$, and $k = o(n^{2/3})$, then $k^4/n^3 = o(1)$. Therefore

$$\exp(nh(p)) = 2^n \exp\left(-\frac{2k^2}{n}\right)(1 + o(1)).$$

For the prefactor,

$$p(1-p) = (1/2 + \delta)(1/2 - \delta) = 1/4 - \delta^2.$$

So

$$\frac{1}{\sqrt{2\pi np(1-p)}} = \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{n}} \frac{1}{\sqrt{1 - 4\delta^2}} = \left(\sqrt{\frac{2}{\pi}} + o(1)\right) \frac{1}{\sqrt{n}}$$

because $\delta \to 0$. Combine them together gives the desired form. □

1.3 Eigenvalues and sums of Hermitian matrices

1.3.1 Proof of the spectral theorem

Exercise 1.3.1 Suppose that the eigenvalues $\lambda_1(A) > \cdots > \lambda_n(A)$ of an $n \times n$ Hermitian matrix are distinct. Show that the associated eigenbasis $u_1(A), \ldots, u_n(A)$ is unique up to rotating each individual eigenvector $u_j(A)$ by a complex phase $e^{i\theta_j}$. In particular, the *spectral projections* $P_j(A) := u_j(A)u_j(A)^*$ are unique. What happens when there is eigenvalue multiplicity.

Solution. *Proof.* Fix j. Since λ_j is an eigenvalue, the eigenspace is $E_j = \ker(A - \lambda_j I)$. Since the eigenvalues are distinct, the characteristic polynomial has λ_j as a simple root, so $\dim E_j = 1$. So any two nonzero eigenvectors $u, v \in E_j$ must be $u = cv$ for some $c \in \mathbb{C} \setminus \{0\}$. If they are unit vectors, then

$$1 = \|u\| = \|cv\| = |c|\|v\| = |c|$$

so $c = e^{e^{i\theta_j}}$ for some $\theta_j \in \mathbb{R}$.

If $u_j' = e^{i\theta_j} u_j$ then the spectral projections

$$P_j(A)' = u_j' u_j'^* = (e^{i\theta_j} u_j)(e^{-i\theta_j} u_j^*) = u_j u_j^* = P_j(A).$$

□

1.3.2 Minimax formulae

Exercise 1.3.2 Show that the inqualities (1.60) are equivalent to the assertion that the diagonal entries $\operatorname{diag}(A) = (a_{11}, a_{22}, \ldots, a_{nn})$ lies in the *permutahedron* of $\lambda_1(A), \ldots, \lambda_n(A)$, defined as the convex hull of the $n!$ permutations of $(\lambda_1(A), \ldots, \lambda_n(A))$ in $\mathbb{R}^n$.

Solution. *Proof.* For hermitian matrices, the diagonals are reals. For any subset of real numbers, the convex hull is just the interval between the maximum and the minimum.

Let $\lambda = (\lambda_1, \ldots, \lambda_n)$. Let $d = \operatorname{diag}(A)$. Define

$$\mathcal{P}(\lambda) := \operatorname{conv}\{\pi(\lambda) : \pi \in S_n\} \subset \mathbb{R}^n.$$

And fix any $S \subset \{1, \ldots, n\}$ of size k. Note that $\sum_{i \in S} d_i \geq \sum_{j=n-k+1}^{n} \lambda_j \iff \sum_{i \notin S} d_i \leq \sum_{j=1}^{n-k} \lambda_j$. So define

$$\mathcal{Q}(\lambda) := \left\{ x \in \mathbb{R}^n : \sum_{i \in S} x_i \leq \sum_{j=1}^{|S|} \lambda_j \quad \forall S, \sum_i x_i = \sum_j \lambda_j \right\}.$$

Then it remains to prove

$$\mathcal{P}(\lambda) = \mathcal{Q}(\lambda).$$

Take a vertex $v = \pi(\lambda)$. For any subset S of size k,

$$\sum_{i \in S} v_i = \sum_{i \in S} \lambda_{\pi(i)} \leq \sum_{i=1}^{k} \lambda_k.$$

Also $\sum_i v_i = \sum_j \lambda_j$. ***why?*** Hence every vertex lies in $\mathcal{Q}(\lambda)$, and by convexity,

$$\mathcal{P}(\lambda) \subset \mathcal{Q}(\lambda).$$

Fix $c \in \mathbb{R}^n$. Let c' be c sorted decreasingly. And let x' denote x sorted decreasingly. We claim that if $x \in \mathcal{Q}(\lambda)$, then

$$c \cdot x \leq \sum_{i=1}^{n} c_i' \lambda_i.$$

Then

$$\max_{x \in \mathcal{Q}(\lambda)} c \cdot x \leq \sum_{i=1}^{n} c_i' \lambda_i$$

Over the permutahedron,

$$\max_{x \in \mathcal{P}(\lambda)} c \cdot x = \max_{\pi \in S_n} c \cdot \pi(\lambda) = \sum_{i=1}^{n} c_i' \lambda_i,$$

because the maximum is attained by matching the largest coordinate of c with the largest λ, etc. Comparing the above two inequalities we have

$$\max_{\mathcal{Q}(\lambda)} c \cdot x \leq \max_{\mathcal{P}(\lambda)} c \cdot x \text{ for all } c.$$

This proves $\mathcal{Q}(\lambda) \subset \mathcal{P}(\lambda)$.

Proof of claim: write c' in difference form: $\alpha_k := c'_k - c'_{k+1} \geq 0$ where $c'_{n+1} := 0$. Then $c'_i = \alpha_i + \cdots + \alpha_n$, and for any vector y,

$$\sum_{i=1}^{n} c'_i y'_i = \sum_{k=1}^{n} \alpha_k \left(\sum_{i=1}^{k} y'_i \right).$$

Now for $x \in \mathcal{Q}(\lambda)$, the upper bound in the inequalities implies the top k partial sum bound:

$$\sum_{i=1}^{k} x'_i = \max_{|S|=k} \sum_{i \in S} x_i \leq \sum_{j=1}^{k} \lambda_j.$$

Plug it in into the preceding equality with $y = x$:

$$\sum_{i=1}^{n} c'_i x'_i \leq \sum_{k=1}^{n} \alpha_k \left(\sum_{j=1}^{k} \lambda_j \right) = \sum_{i=1}^{n} c'_i \lambda_i.$$

Finally $c \cdot x \leq \sum_i c'_i x'_i$. Claim proved. □

Exercise 1.3.3 (Wielandt minimax formula). Let $1 \leq i_1 < \cdots < i_k \leq n$ be integers. Define a *partial flag* to be a nested collection $V_1 \subset \cdots \subset V_k$ of subspaces of $\mathbb{C}^n$ such that $\dim(V_j) = i_j$ for all $1 \leq j \leq k$. Define the associated *Schubert variety* $X(V_1, \ldots, V_k)$ to the the collection of all k-dimensional subspaces W such that $\dim(W \cap V_j) \geq j$. Show that for any $n \times n$ matrix A,

$$\lambda_{i_1}(A) + \cdots + \lambda_{i_k}(A) = \sup_{V_1, \ldots, V-k} \inf_{W \in X(V_1, \ldots, V_k)} \operatorname{tr}(A \lfloor_W).$$

1.3.3 Eigenvalue inequalities

Exercise 1.3.4. Verify the inequalities (1.63) and (1.55) by hand in the case when A and B commute (and are thus simultaneously diagonalisable), without the use of minimax formulae.

Solution. *Proof.* We use the fact that sum of any k entries $\leq$ sum of the k largest entries. Assume A and B commute, then they are simultaneously diagonalisable, which means there exists a unitary U and real vectors

$$\alpha = (\alpha_1, \ldots, \alpha_n), \quad \beta = (\beta_1, \ldots, \beta_n)$$

such that

$$U^* A U = \operatorname{diag}(\alpha_1, \ldots, \alpha_n), \quad U^* B U = \operatorname{diag}(\beta_1, \ldots, \beta_n)$$

Hence

$$U^*(A + B)U = \operatorname{diag}(\alpha_1 + \beta_1, \ldots, \alpha_n + \beta_n).$$

Let a and b be α and β in the decreasing order, and let

$$c = (c_1, \ldots, c_n)$$

be the decreasing arrangement of $\{\alpha_j + \beta_j\}_{j=1}^n$. Fix the indices $1 \leq i_1 < \cdots < i_k \leq n$. Let $T \subset \{1, \ldots, n\}$ be the set of positions corresponding to the eigenvalues $\lambda_{i_1}(A + B), \ldots, \lambda_{i_k}(A + B)$, i.e. T indicates the k diagonal entries $\alpha_j + \beta_j$ whose decreasing rearrangement entries are $c_{i_1}, \ldots, c_{i_k}$. Then

$$\sum_{r=1}^{k} \lambda_{i_r}(A + B) = \sum_{r=1}^{k} c_{i_r} = \sum_{j \in T} (\alpha_j + \beta_j) = \sum_{j \in T} \alpha_j + \sum_{j \in T} \beta_j.$$

Then we bound each part. First we notice

$$\sum_{j\in T}\beta_j \le \sum_{r=1}^{k} b_r.$$

For the A-part, notice that $j_r \le i_r$ (cannot be further pushed forward), the maximum of $\sum_{r=1}^{k} a_{j_r}$ is attained at $j_r = i_r$. So

$$\sum_{j\in T}\alpha_j \le \sum_{r=1}^{k} a_{i_r}.$$

Then

$$\sum_{r=1}^{k}\lambda_{i_r}(A+B) = \sum_{r=1}^{k} c_{i_r} \le \sum_{r=1}^{k} a_{i_r} + \sum_{r=1}^{k} b_r = \sum_{r=1}^{k}\lambda_{i_r}(A) + \sum_{r=1}^{k}\lambda_r(B).$$

A similar argument works for the Weyl inequality. □

Exercise 1.3.5. Establish the dual Lidskii inequality

$$\begin{aligned}\lambda_{i_1}(A+B)+\cdots+\lambda_{i_k}(A+B) \ge &\lambda_{i_1}(A)+\cdots+\lambda_{i_k}(A)\\ &+\lambda_{n-k+1}(B)+\cdots+\lambda_n(B)\end{aligned}$$

for any $1 \le i_1 < \cdots < i_k \le n$ and the dual Weyl inequality

$$\lambda_{i+j-n}(A+B) \ge \lambda_i(A) + \lambda_j(B)$$

whenever $1 \le i, k, i+j-n \le n$.

Solution. *Proof.* Replace the A and B in the Lidskii inequality with $-A$ and $-B$:

$$\lambda_{i_1}(-A-B)+\cdots+\lambda_{i_k}(-A-B) \le \lambda_{i_1}(-A)+\cdots+\lambda_{i_k}(-A)+\lambda_1(-B)+\cdots+\lambda_k(-B).$$

Since $\lambda_j(-M) = -\lambda_{n-j+1}(M)$, we have

$$\sum_{t=1}^{k}\lambda_{n-i_t+1}(A+B) \ge \sum_{t+1}^{k}\lambda_{n-i_t+1}(A) + \sum_{t=1}^{k}\lambda_{n-t+1}(B).$$

Relabeling the indices will give the desired form. A similar argument works for the Weyl inequality. □

Exercise 1.3.6. Use the Lidskii inequality to establish the more general inequality

$$\sum_{i=1}^{n} c_i\lambda_i(A+B) \le \sum_{i=1}^{n} c_i\lambda_i(A) + \sum_{i=1}^{n} c_i^*\lambda_i(B)$$

whenever $c_1,\ldots,c_n \ge 0$, and $c_1^* \ge \ldots \ge c_n^* \ge 0$ is the decreasing rearrangement of $c_1,\ldots,c_n$. (*Hint:* express c_i as the integral of $\mathbb{1}(c_i \ge \lambda)$ as λ runs from 0 to infinity. For each fixed λ, apply (1.63).) Combine this with Hölders inequality to conclude the p-Weilandt-Hoffman inequality

$$\|(\lambda_i(A+B)-\lambda_i(A))_{i=1}^n\|_{\ell_n^p} \le \|B\|_{S^p}$$

for any $1 \le p \le \infty$, where

$$\|(a_i)_{i=1}^n\|_{\ell_n^p} := (\sum_{i=1}^{n}|a_i|^p)^{1/p}$$

is the usual ℓ^p norm (with the usual convention that $\|(a_i)_{i=1}^n\|_{\ell_n^\infty} := \sup_{1\le i\le p}|a_i|$), and

$$\|B\|_{S^p} := \|(\lambda_i(B))_{i=1}^n\|_{\ell_n^p}$$

is the p-Schatten norm of B.

Solution. *Proof.* For each c_i, write it as

$$c_i = \int_0^\infty \mathbb{1}_{(c_i \geq \lambda)} \, d\lambda.$$

Then

$$\sum_{i=1}^n c_i \lambda_i(A) = \sum_{i=1}^n \left(\int_0^\infty \mathbb{1}_{(c_i \geq \lambda)} \, d\lambda \right) \lambda_i(A) = \int_0^\infty \sum_{i: c_i \geq \lambda} \lambda_i(A) \, d\lambda.$$

Let $S_\lambda := \{i : c_i \geq \lambda\}$. By Lidskii inequality,

$$\sum_{i \in S_\lambda} \lambda_i(A+B) \leq \sum_{i \in S_\lambda} \lambda_i(A) + \sum_{j=1}^{|S_\lambda|} \lambda_i(B).$$

Integrate both sides gives

$$\sum_{i=1}^n c_i \lambda_i(A+B) \leq \sum_{i=1}^n c_i \lambda_i(A) + \int_0^\infty \sum_{j=1}^{|S_\lambda|} \lambda_i(B).$$

Note that

$$\int_0^\infty \sum_{j=1}^{|S_\lambda|} \lambda_i(B) \, d\lambda = \int_0^\infty \sum_{j=1}^n \mathbb{1}_{(|S_\lambda| \geq j)} \lambda_i(B) \, d\lambda = \sum_{j=1}^n \left(\int_0^\infty \mathbb{1}_{(|S_\lambda| \geq j)} \, d\lambda \right) \lambda_i(B).$$

Note that $|S_\lambda| \geq j$ means that j-th largest c_i is $\geq \lambda$, i.e. $c_j^* \geq \lambda$. So

$$\int_0^\infty \mathbb{1}_{(|S_\lambda| \geq j)} \, d\lambda = \int_0^\infty \mathbb{1}_{(c_j^* \geq \lambda)} \, d\lambda = c_j^*$$

hence

$$\int_0^\infty \sum_{j=1}^{|S_\lambda|} \lambda_i(B) \, d\lambda = \sum_{j=1}^n c_j^* \lambda_i(B).$$

This proves the generalized Lidskii inequality.

Let

$$d_i := \lambda_i(A+B) - \lambda_i(A).$$

From above we have

$$\sum_{i=1}^n c_i d_i \leq \sum_{i=1}^n c_i^* \lambda_i(B).$$

For any $1 \leq p < \infty$ and its conjugate exponent q,

$$\sum_{i=1}^n c_i^* \lambda_i(B) \leq \left(\sum_{i=1}^n |c_i^*|^q \right)^{1/q} \left(\sum_{i=1}^n |\lambda_i(B)|^p \right)^{1/p} = \|c\|_{\ell_n}^q \|B\|_{S^p}.$$

Taking the supremum over all $c \geq 0$ with $\|c\|_{\ell_n}^q = 1$ (dual characterization of the ℓ^p norm) yields the desired inequality. □

Exercise 1.3.7. Show that the p-Schatten norms are indeed norms on the space of hermitian matrices for every $1 \leq p \leq \infty$.

Solution. *Proof.* Positivity is immediate by ℓ^p-norm. If $\|A\|_{S^p} = 0$, then all eigenvalues are 0, hence $A = 0$.

Since $\lambda_i(cA) = |c|\lambda_i(A)$ for all $c \in \mathbb{R}$ (since cA being Hermitian forces $c \in \mathbb{R}$, so

$$\|cA\|_{S^p} = \begin{cases} \left(\sum_{i=1}^n ||c|\lambda_i|^p\right)^{1/p} = |c| \left(\sum_{i=1}^n |\lambda_i|^p\right)^{1/p} = |c|\|A\|_{S^p}, & 1 \le p < \infty, \\ \sup_{1\le i\le p} ||c|\lambda_i| = |c| \sup_{1\le i\le p} |\lambda_i|, & p = \infty. \end{cases}.$$

For the triangular inequality, we apply the Ky Fan ineuality

$$\sum_{i=1}^k \lambda_i(A+B) \le \sum_i^k (\lambda_i(A) + \lambda_i(B)).$$

So by the Schur-convexity and Minkowski inequality,

$$\|A+B\|_{S^p} = \|\lambda(A+B)\|_{\ell^p} \le \|\lambda(A) + \lambda(B)\| \le \|\lambda(A)\|_{\ell^p} + \|\lambda(B)\|_{\ell^p} = \|A\|_{S^p} + \|B\|_{S^p}.$$

For $p = \infty$,

$$\|A+B\|_{S^\infty} = \|A+B\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}} + \|B\|_{\mathrm{op}} = \|A\|_{S^\infty} + \|B\|_{S^\infty}.$$

□

Exercise 1.3.8. Show that for any $1 \le p \le \infty$ and any Hermitian matrix $A = (a_{ij})_{1\le i,j\le n}$, one has

$$\|(a_{ii})_{i=1}^n\|_{\ell_n^p} \le \|A\|_{S^p}.$$

Solution. *Proof.* Since A is Hermitian, we can diagonalize A:

$$A = UDU^*$$

where D is the diagonal matrix of the eigenvalues of A and U is unitary. Then the i-th diagonal entry is

$$a_{ii} = e_i^* A e_i = e_i U D U^* e_i = \sum_{j=1}^n |u_{ij}|^2 \lambda_j$$

where u_{ij} is the entry of U. Since U is unitary,

$$\sum_{j=1}^n |u_{ij}|^2 = \sum_{i=1}^n |u_{ij}|^2 = 1.$$

For $p \ge 1$, $\phi(x) = |x|^p$ is convex. Applying Jensen's inequality to this we have

$$\sum_{i=1}^n |a_{ii}|^2 \le \sum_{i=1}^n \sum_{j=1}^n |u_{ij}|^2 |\lambda_j|^p = \sum_{j=1}^n |\lambda_j|^p \left(\sum_{i=1}^n |u_{ij}|^2\right) = \sum_{j=1}^n |\lambda_j|^p.$$

□

Exercise 1.3.9. Establish the Hölder inequality

$$|\operatorname{tr}(AB)| \le \|A\|_{S^p} \|B\|_{S^{p'}}$$

whenever $1 \le p, p' \le \infty$ with $1/p + 1/p' = 1$, and A, B are $n \times n$ Hermitian matrices. (*Hint:* Diagnolise one of the matrices and use the preceding exercise.)

Solution. *Proof.* Let $\{u_i\}_{i=1}^n$ be an orthonormal eigenbasis of B, then $Bu_i = \lambda_i(B)u_i$. Then

$$\operatorname{tr}(AB) = \sum_{i=1}^n u_i^* A B u_i = \sum_{i=1}^n \lambda_i(B) u_i^* A u_i,$$

hence

$$|\operatorname{tr}(AB)| \leq \sum_{i=1}^{n} |\lambda_i(B)||u_i^* A u_i|.$$

Applying the Hölder inequality,

$$\sum_{i=1}^{n} |\lambda_i(B)||u_i^* A u_i| \leq \|(|\lambda_i(B)|)_{n=1}^n\|_{\ell^{p'}} \|(|u_i^* A u_i|)_{i=1}^n\|_{\ell^p} = \|B\|_{S^{p'}} \|A\|_{S^p}.$$

The first part is by definition and the second part is by the preceding exercise. □

1.3.4 Eigenvalue deformation

Exercise 1.3.10 (Dimension count). Suppose that $n \geq 2$. Show that the space of Hermitian matrices with at least one repeated eigenvalue has codimension 3 in the space of all Hermitian matrices, and the space of real symmetric matrices with at least one repeated eigenvalue has codimension 2 in the space of all real symmetric matrices. (When $n = 1$, repeated eigenvalues of course do not occur.)

Exercise 1.3.11 Use a similar argument to the one above to establish (1.68) without using minimax formulae or Lidskiis inequality.

Solution. *Proof.* Define

$$\lambda_i(t) = \lambda_i(A(t)) \quad \text{where} \quad A(t) = A + tB.$$

Apply the Hardamard first variation formula, we see that

$$\dot{\lambda}_i(t) = u_i(t)^* B u_i(t).$$

Integrating both sides and taking summation of squares gives

$$\sum_{i=1}^{n} \left| \int_0^1 \dot{\lambda}_i(t)\ dt \right|^2 = \sum_{i=1}^{n} \left| \int_0^1 u_i(t)^* B u_i(t)\ dt \right|^2 \leq \sum_{i=1}^{n} \int_0^1 |u_i(t)^* B u_i(t)|^2\ dt \leq \int_0^1 \sum_{i=1}^{n} |\lambda_i(B)|^2\ dt = \|B\|_F^2.$$

The left hand side is exactly $\sum_{i=1}^n |\lambda_i(A+B) - \lambda_i(A)|^2$ by then fundamental theorem of calculus. □

Exercise 1.3.12 Use a similar argument to the one above to deduce Lidskiis inequality (1.63) from Proposition 1.3.4 rather than Exercise 1.3.3.

Solution. *Proof.* Define

$$\lambda_m(t) = \lambda_m(A(t)) \quad \text{where} \quad A(t) = A + tB.$$

And

$$F(t) := \sum_{j=1}^{k} \lambda_{i_j}(t).$$

Then

$$F'(t) = \sum_{j=1}^{k} \dot{\lambda}_{i_j}(t) = \sum_{j=1}^{k} u_{i_j}(t)^* B u_{i_j}(t) = \operatorname{tr}(B \restriction_W) \leq \sum_{j=1}^{k} \lambda_j(B).$$

where

$$W = \operatorname{span}(u_{i_1}(t), u_{i_2}(t), \ldots, u_{i_k}(t)).$$

Integrating both sides gives

$$\sum_{j=1}^{k} \lambda_{i_j}(A+B) - \sum_{j=1}^{k} \lambda_{i_j}(A) = F(1) - F(0) = \int_0^1 F'(t)\ dt = \int_0^1 \sum_{j=1}^{k} \lambda_j(B)\ dt = \sum_{j=1}^{k} \lambda_j(B).$$

□

Exercise 1.3.13. Suppose that $A = A(t)$ depends smoothly on t. By differentiating (1.71) and (1.72), establish the *Hadamard second variation formula*

$$\frac{d^2}{dt^2}\lambda_k = u_k^* \ddot{A} u_k + 2\sum_{j\neq k} \frac{|u_j^* \dot{A} u_k|^2}{\lambda_k - \lambda_j}$$

whenever A has simple spectrum and $1 \leq k \leq n$.

Solution. *Proof.* Starting from (1.71), taking derivative of it, we obtain

$$\ddot{A}u_i + 2\dot{A}\dot{u}_i + A\ddot{u}_i = \ddot{\lambda}_i u_i + 2\dot{\lambda}_i \dot{u}_i + \lambda_i \ddot{u}_i.$$

Taking inner products of it with u_i. Using $\dot{u}_i \perp u_i$ and $u_i^* A = \lambda_i u_i^*$, cancels the $A\ddot{u}_i$ and $\lambda_i \ddot{u}_i$ terms, we have

$$u_i^* \ddot{A} u_i + 2u_i^* \dot{A}\dot{u}_i = \ddot{\lambda}_i.$$

Since $\{u_j\}_{j=1}^n$ forms a orthonormal basis, and $\dot{u}_i \perp u_i$, we can expand $\dot{u}_i$ as

$$\dot{u}_i = \sum_{j=1}^n (u_j^* \dot{u}_i) u_j = \sum_{j\neq i} (u_j^* \dot{u}_i) u_j.$$

So

$$u_i^* \dot{A}\dot{u}_i = u_i^* \dot{A} \sum_{j\neq i} (u_j^* \dot{u}_i) u_j = \sum_{j\neq i} (u_j^* \dot{u}_i) u_i^* \dot{A} u_j$$

since each $u_j^* \dot{u}_i$ is a constant. Now we take inner product of (1.71) with u_j for $j \neq i$, we have

$$u_j^* \dot{A} u_i + \lambda_j u_j^* \dot{u}_i = \lambda_i u_j^* \dot{u}_i$$

so

$$u_j^* \dot{u}_i = \frac{u_j^* \dot{A} u_i}{\lambda_i - \lambda_j}.$$

Therefore

$$u_i^* \dot{A}\dot{u}_i = \sum_{j\neq i} \frac{u_j^* \dot{A} u_i}{\lambda_i - \lambda_j} u_i^* \dot{A} u_j = \sum_{j\neq i} \frac{(u_j^* \dot{A} u_i)\overline{(u_j^* \dot{A} u_i)}}{\lambda_i - \lambda_j} = \sum_{j\neq i} \frac{|u_j^* \dot{A} u_i|^2}{\lambda_i - \lambda_j}$$

thus

$$\ddot{\lambda}_i = u_i^* \ddot{A} u_i + 2\sum_{j\neq i} \frac{|u_j^* \dot{A} u_i|^2}{\lambda_i - \lambda_j}.$$

□

1.3.5 Minors

Exercise 1.4.14 (Cauchy interlacing law). For any $n \times n$ Hermitian matrix A_n with top left $n-1 \times n-1$ minor A_{n-1}, then

$$\lambda_{i+1}(A_n) \leq \lambda_i(A_{n-1}) \leq \lambda_i(A_n)$$

for all $1 \leq i < n$. Show furthermore that the space of A_n for which equality holds in one of the inequalities above has codimension 2 (for Hermitian matrices) or 1 (for real symmetric matrices).

Solution. *Proof.* Define

$$H = \{\tilde{y} = (y, 0) \in \mathbb{C}^n : y \in \mathbb{C}^{n-1}\}.$$

Then $\dim(H) = n-1$. For any $y \in \mathbb{C}^{n-1}$, $\tilde{y} = (y, 0) \in \mathbb{C}^n$ and

$$\tilde{y}^* A_n \tilde{y} = y^* A_{n-1} y. \tag{1}$$

Let $S \leq \mathbb{C}^n$ with $\dim(S) = i+1$. Then $\dim(S \cap H) \geq i$. To see this, H excludes one direction, so at most one direction of S can lie outside of H, i.e., at least i directions would be in H. Or noticing that $H + S \subset \mathbb{C}^n$ so $\dim(H+S) \leq n$. Then

$$\dim(H \cap S) = \dim(H) + \dim(S) - \dim(H+S) \geq (n-1) + (i+1) - n = i.$$

Pick an i-dimensional subspace $W \subset S \cap H$. Since $W \subset S$,

$$\inf_{v \in S, |v|=1} v^* A_n v \leq \inf_{v \in W, |v|=1} v^* A_n v.$$

Since $W \subset H$, every $v \in W$ is in the form $v = \tilde{w} = (w, 0)$ where $w \in \mathbb{C}^{n-1}$. Applying (1) we have

$$\inf_{v \in W, |v|=1} v^* A_n v = \inf_{w \in W', |w|=1} w^* A_{n-1} w$$

where W' is the corresponding i-dimensional subspace of $\mathbb{C}^{n-1}$. By the Courant-Fischer minimax theorem,

$$\inf_{v \in S, |v|=1} v^* A_n v \leq \inf_{w \in W', |w|=1} w^* A_{n-1} w \leq \lambda_i(A_{n-1}).$$

Taking supremum over all $(i+1)$-dimensional subspaces of $\mathbb{C}^n$, we have

$$\lambda_{i+1}(A_n) \leq \lambda_i(A_{n-1}).$$

The other part is similar. Let $S \leq \mathbb{C}^n$ with $\dim(S) = n-i+1$. Then $\dim(S \cap H) \geq n-i$. To see this, noticing that $H + S \subset \mathbb{C}^n$ so $\dim(H+S) \leq n$. Then

$$\dim(H \cap S) = \dim(H) + \dim(S) - \dim(H+S) \geq (n-1) + (n-i+1) - n = n-i.$$

Pick an $(n-i)$-dimensional subspace $W \subset S \cap H$. Since $W \subset S$,

$$\sup_{v \in W, |v|=1} v^* A_n v \leq \sup_{v \in S, |v|=1} v^* A_n v.$$

Since $W \subset H$, every $v \in W$ is in the form $v = \tilde{w} = (w, 0)$ where $w \in \mathbb{C}^{n-1}$. Applying (1) we have

$$\sup_{v \in W, |v|=1} v^* A_n v = \sup_{w \in W', |w|=1} w^* A_{n-1} w$$

where W' is the corresponding $(n-i)$-dimensional subspace of $\mathbb{C}^{n-1}$. By the Courant-Fischer minimax theorem,

$$\lambda_i(A_{n-1}) \leq \sup_{w \in W', |w|=1} w^* A_{n-1} w \leq \sup_{v \in S, |v|=1} v^* A_n v.$$

Taking infimum over all $(n-i+1)$-dimensional subspaces of $\mathbb{C}^n$, we have

$$\lambda_i(A_{n-1}) \leq \lambda_i(A_n).$$

□

2 Random matrices

2.1 Concentration of measure

2.1.1 Linear combinations, and the moment method.

Exercise 2.1.1. Show that the $O(t^2(b-a)^2)$ factor in (2.10) can be replaced with $t^2(b-a)^2/8$, and that this is sharp. (*Hint:* Use Jensen's inequality, Exercise 1.1.8.)

Proof. Since $x \mapsto e^{tx}$ is convex, the line adjacent any two points (a, e^{ta}), (b, e^{tb}) should be bounded below by the curve e^{tx}, i.e.,

$$e^{tx} \leq L(x) = \frac{b-x}{b-a}e^{ta} + \frac{x-a}{b-a}e^{tb}.$$

Letting $x = X$ and taking expectations gives

$$\mathbb{E}e^{tX} \leq \frac{b}{b-a}e^{ta} - \frac{a}{b-a}e^{tb}.$$

Let $p = -\frac{a}{b-a}$ and $u = t(b-a)$. Then we have

$$\mathbb{E}e^{tX} \leq (1-p)e^{-pu} + pe^{(1-p)u} = e^{-pu}(1-p+pe^u) = \exp(-pu + \log(1-p+pe^u)).$$

Let

$$\phi(u) = -pu + \log(1-p+pe^u).$$

It suffices to show $\phi(u) \leq u^2/8$. Taking derivative

$$\phi'(u) = -p + \frac{pe^u}{1-p+pe^u}.$$

Letting $\phi'(u) = 0$ we have $u = 0$. Then consider the second derivative

$$\phi''(u) = \frac{p(1-p)e^u}{(1-p+pe^u)^2}.$$

If we let $v = \frac{pe^u}{1-p+pe^u} \in [0,1]$ then $\phi''(u) = v(1-v) \leq 1/4$. Therefore by Taylor expansion around 0, for some $z \in (0,u)$, we have

$$\phi(u) = \phi(0) + u\phi'(0) + \frac{u^2}{2}\phi''(u) \leq 0 + 0 + \frac{u^2}{2} \cdot \frac{1}{4} = \frac{u^2}{8}.$$

□

Exercise 2.1.2. Let $0 < p < 1/2$ be fixed independently of n, and let $X_1, \ldots, X_n$ be iid copies of a Bernoulli random variable that equals 1 with probability p, thus $\mu_i = p$ and $\sigma_i^2 = p(1-p)$, and so $\mu = np$ and $\sigma^2 = np(1-p)$. Using Stirlings formula, show that

$$\mathbb{P}(|S_n - \mu| \geq \lambda\sigma) \geq c\exp(-C\lambda^2)$$

for some absolute constants $C, c > 0$ and all $\lambda \leq c\sigma$. What happens when λ is much larger than σ?

Proof. Firstly

$$\mathbb{P}(|S_n - \mu| \geq \lambda\sigma) \geq \mathbb{P}(S_n \geq np + \lambda\sigma).$$

Let $k = \lceil np + \lambda\sigma \rceil$, then

$$\mathbb{P}(S_n \geq np + \lambda\sigma) \geq \mathbb{P}(S_n = k) = \binom{n}{k}p^k(1-p)^{n-k}.$$

Let $\gamma = k/n$, then by Stirling's formula, or more precisely the Entropy formula,

$$\binom{n}{k}p^k(1-p)^k = \exp(nh(\gamma) + o(n))p^k(1-p)^{n-k} = \exp\left(n(h(\gamma) + \gamma\log p + (1-\gamma)\log(1-p)) + o(n)\right)$$

where $h(\gamma)$ is the entropy function. Let

$$f(\gamma) = h(\gamma) + \gamma\log p + (1-\gamma)\log(1-p).$$

Then

$$\mathbb{P}(S_n = k) = \exp(nf(\gamma) + o(n)).$$

We use Taylor series to bound the exponent. First we have

$$f'(\gamma) = h'(\gamma) + \log p - \log(1-p) = \log\left(\frac{1-\gamma}{\gamma}\frac{p}{1-p}\right).$$

$f'(\gamma) = 0$ gives $\gamma = p$.

$$f''(\gamma) = -\frac{1}{\gamma(1-\gamma)}.$$

So by Taylor expansion around p, we have

$$f(\gamma) = f(p) + f'(p)(\gamma - p) + f''(p)\frac{(\gamma-p)^2}{2} + O((\gamma-p)^3) = -\frac{(\gamma-p)^2}{2p(1-p)} + O((\gamma-p)^3).$$

Since $k = \lceil np + \lambda\sigma \rceil$ and $\sigma^2 = np(1-p)$, we have

$$\gamma - p = \frac{k - np}{n} = \frac{\lambda\sigma}{n} = \lambda\sqrt{\frac{p(1-p)}{n}}$$

and therefore

$$nf(\gamma) = -\frac{\lambda^2}{2} + O\left(\frac{\lambda^3}{\sqrt{n}}\right).$$

If $\lambda \le c\sigma$,

$$\frac{\lambda^3}{\sqrt{n}} \le \frac{c\sigma}{\sqrt{n}}\lambda^2 = c\sqrt{p(1-p)}\lambda^2 < \frac{c}{2}\lambda^2$$

by optimizing in t. So

$$\exp(nf(\gamma) + o(n)) > \exp\left(-\frac{1}{2}(1-c)\lambda^2 + o(n)\right).$$

(Dropping the ceiling in expressing $\gamma - p$ is possible since changing k by ± 1 changes γ by $\pm 1/n$; in fact

$$\log\frac{\mathbb{P}(S_n = k+1)}{\mathbb{P}(S_n = k)} = \log\left(\frac{n-k}{k+1}\frac{p}{1-p}\right) = O\left(\frac{\lambda}{\sigma}\right).$$

So for $\lambda \le c\sigma$ this ratio is bounded between two absolute constants. Therefore dropping the ceiling only changes probabilities by a multiplicative absolute constant.)

But we see that there's still an $o(n)$ left. So we cannot use the entropy formula directly. Instead we will modify it. From the Stirling's formula we have

$$\log(n!) = n\log n - n + \frac{1}{2}\log(2\pi n) + o(1)$$

since $\log(1 + o(1)) = o(1)$. Now apply this to $\binom{n}{m}$ and write $m = \gamma n$, $0 < \gamma < 1$, we have

$$\log\binom{n}{m} = \log(n!) - \log(m!) - \log((n-m)!) = nh(\gamma) - \frac{1}{2}\log(2\pi n\gamma(1-\gamma)) + o(1)$$

where $h(\gamma)$ is the entropy formula. Exponentiating gives

$$\binom{n}{M} = (1 + o(1))\frac{1}{\sqrt{2\pi n\gamma(1-\gamma)}}\exp(nh(\gamma)).$$

When λ is much larger than n, $\lambda\sigma$ is much larger than n, so

$$\mathbb{P}(S_n \ge \mu + \lambda\sigma) = 0.$$

□

Exercise 2.1.3. Show that the term $\exp(-c\lambda\sigma/K)$ in (2.11) can be replaced with $(\lambda K/\sigma)^{-c\lambda\sigma/K}$ (which is superior when $\lambda K \gg \sigma$). (Hint: Allow t to exceed 1.) Compare this with the results of Exercise 2.1.2.

Proof. Subtract means so $\mu_i = 0$ and hence $\mu = 0$. Divide by K gives $|X_i| \leq 1$. By Taylor expansion

$$e^{tX} = 1 + tX + \sum_{k\geq 2} \frac{(tx)^k}{k!} \leq 1 + tx + x^2 \sum_{k\geq 2} \frac{t^k}{k!} = 1 + tx + (e^t - 1 - t)x^2.$$

Taking expectation and letting $x = X$ gives

$$\mathbb{E}e^{tX} \leq 1 + (e^t - 1 - t)\mathbb{E}X^2 = 1 + (e^t - 1 - t)\sigma^2 \leq \exp((e^t - 1 - t)\sigma^2).$$

Applying this to each $X = X_i$ and we have

$$\mathbb{E}e^{tS_n} = \prod_{i=1}^n \mathbb{E}e^{tX_i} \leq \exp\left((e^t - 1 - t)\sum_{i=1}^n \sigma_i^2\right) = \exp((e^t - 1 - t)\sigma^2).$$

By Markov's inequality,

$$\mathbb{P}(S_n \geq \lambda\sigma) \leq \exp((e^t - 1 - t)\sigma^2 - t\lambda\sigma).$$

It suffices to optimize

$$\phi(t) = (e^t - 1 - t)\sigma^2 - \lambda\sigma t.$$

Taking derivative

$$\phi'(t) = \sigma^2\left(e^t - 1 - \frac{\lambda}{\sigma}\right).$$

So $\phi'(t) = 0$ when $t = \log\left(1 + \frac{\lambda}{\sigma}\right)$. And we can see that

$$\phi(t) \leq \lambda\sigma - \sigma(\sigma + \lambda)\log\left(1 + \frac{\lambda}{\sigma}\right).$$

When $\lambda \gg \sigma$, the dominate term in $\phi(t)$ is $-\sigma\lambda\log(\lambda/\sigma)$. Exponentiating,

$$\exp(\phi(t)) \leq \left(\frac{1 + \lambda/\sigma}{e}\right)^{-\sigma\lambda} \leq \left(\frac{\lambda}{\sigma}\right)^{-c\sigma\lambda}$$

when $\lambda \geq (e-1)^{-1}\sigma$. □

Exercise 2.1.4. (Hoeffding's inequality). Let $X_1, \ldots, X_n$ be independent real variables, with X_i taking values in an interval $[a_i, b_i]$, and let $S_n := X_1 + \ldots + X_n$. Show that one has

$$\mathbb{P}(|S_n - \mathbb{E}S_n| \geq \lambda\sigma) \leq C\exp(-c\lambda^2)$$

for some absolute constants $C, c > 0$, where $\sigma^2 := \sum_{i=1}^n |b_i - a_i|^2$.

Proof. Let

$$Y_i = X_i - \mathbb{E}X_i.$$

Then $\mathbb{E}Y_i = 0$ and $S_n - \mathbb{E}S_n = \sum_{i=1}^n Y_i$. Therefore

$$\mathbb{P}(|S_n - \mathbb{E}S_n| \geq \lambda\sigma) = \mathbb{P}\left(\left|\sum_{i=1}^n Y_i\right| \geq \lambda\sigma\right) \leq \mathbb{P}\left(\sum_{i=1}^n Y_i \geq \lambda\sigma\right) + \mathbb{P}\left(\sum_{i=1}^n Y_i \leq -\lambda\sigma\right)$$

By Markov's inequality and Exercise 2.1.1.,

$$\mathbb{P}\left(\sum_{i=1}^n Y_i \geq \lambda\sigma\right) \leq \exp(-t\lambda\sigma)\prod_{i=1}^n \mathbb{E}\exp(tY_i) \leq \exp\left(\sum_{i=1}^n \frac{t^2(b_i - a_i)^2}{8} - t\lambda\sigma\right) = \exp\left(\frac{\sigma^2}{8}t^2 - \lambda\sigma t\right).$$

Applying the same bound to $-Y_i$ we get the lower tail

$$\mathbb{P}\left(\sum_{i=1}^{n} Y_i \leq -\lambda\sigma\right) \leq \exp\left(\frac{\sigma^2}{8}t^2 - \lambda\sigma t\right).$$

Combine them together

$$\mathbb{P}(|S_n - \mathbb{E}S_n| \geq \lambda\sigma) \leq 2\exp\left(\frac{\sigma^2}{8}t^2 - \lambda\sigma t\right).$$

Optimizing the exponent in t, the minimum occurs when $t = \frac{4\lambda}{\sigma}$,

$$\exp\left(\frac{\sigma^2}{8}t^2 - \lambda\sigma t\right) \geq -2\lambda^2.$$

□

Exercise 2.1.5. Suppose we replace the hypothesis $|X_i| \leq 1$ in Azuma's inequality with the more general hypothesis $|X_i| \leq c_i$ for some scalars $c_i > 0$. Show that we still have (2.13), but with $\sqrt{n}$ replaced by $(\sum_{i=1}^{n} c_i^2)^{1/2}$.

Proof. We follow the proof of the original Azuma's inequality until applying (2.10) to the conditional expectation, instead we use the refined version in Exercise 2.1.1. and the bound $|X_i| \leq c_i$,

$$\mathbb{E}(\exp(tX_n) \mid X_1, \ldots, X_{n-1}) \leq \exp\left(\frac{t^2}{8}(2c_n)^2\right) = \exp\left(\frac{t^2}{2}c_n^2\right)$$

and

$$\mathbb{E}\exp(tS_n) \leq \exp\left(\frac{t^2}{2}c_n^2\right)\mathbb{E}\exp(tS_{n-1}).$$

Iterating this argument gives

$$\mathbb{E}\exp(tS_n) \leq \exp\left(\frac{t^2}{2}\sum_{i=1}^{n} c_i^2\right).$$

Let $u = \sum_{i=1}^{n} c_i^2$, and by Markov's inequality,

$$\mathbb{P}(S_n \geq \lambda\sqrt{n}) \leq \exp\left(\frac{t^2}{2}u - t\lambda\sqrt{u}\right).$$

Optimizing in t, the minimum of the exponent occurs when $t = \lambda/\sqrt{u}$:

$$\mathbb{P}(S_n \geq \lambda\sqrt{u}) \leq \exp\left(-\frac{1}{2}\lambda^2\right).$$

□

2.1.2 The truncation method.

Exercise 2.1.6. Show that the hypothesis that A is sufficiently large can be replaced by the hypothesis that $A > 0$ is independent of n. *Hint:* There are several approaches available. One can adapt the above proof; one can modify the proof of the Chernoff inequality directly; or one can figure out a way to deduce the small A case from the large A case.

Exercise 2.1.7. Show that the subgaussian hypothesis can be generalised to a sub-exponential tail hypothesis

$$\mathbb{P}(|X| \geq t) \leq C\exp(-ct^p)$$

provided that $p > 1$. Show that the result also extends to the case $0 < p \leq 1$, except with the exponent $\exp(-c_A n)$ replaced by $\exp(-c_A n^{p-\epsilon})$ for some $\epsilon > 0$. (I do not know if the ϵ loss can be removed, but it is easy to see that one cannot hope to do much better than this, just by considering the probability that X_1 (say) is already as large as An.)

2.1.3 Lipschitz combinations

Exercise 2.1.8. Show that McDiarmid's inequality implies Hoeffding's inequality (Exercise 2.1.4).

Proof. Suppose $x_i \in [a_i, b_i]$. Let

$$F(x_1, \ldots, x_n) := \sum_{i=1}^n x_i.$$

Then $F(X) = S_n$ and $\mathbb{E}F(X) = \mathbb{E}S_n$. Then

$$|F(x_1, \ldots, x_i, \ldots, x_n) - F(x_1, \ldots, x_i', \ldots, x_n)| = |x_i - x_i'| \leq b_i - a_i.$$

Then applying McDiarmid we have the Hoeffding's inequality. □

Exercise 2.1.9. Let $X_1, \ldots, X_n$ be independent real gaussian variables with $X_i = N(\mu_i, \sigma_i^2)_{\mathbf{R}}$, and let $c_1, \ldots, c_n$ be real constants. Show that $c_1 X_1 + \ldots + c_n X_n$ is a real gaussian with mean $\sum_{i=1}^n c_i \mu_i$ and variance $\sum_{i=1}^n |c_i|^2 \sigma_i^2$.

Show that the same claims also hold with complex gaussians and complex constants c_i.

Proof. Write

$$X_i = \mu_i + \sigma Z_i, \quad Z_i \sim N(0,1).$$

Then

$$Y_i := \sum_{i=1}^n c_i X_i = \sum_{i=1}^n c_i \mu_i + \sum_{i=1}^n c_i \sigma_i Z_i =: m + \sum_{i=1}^n a_i Z_i$$

where $m = \sum c_i \mu_i$ and $a_i = c_i \sigma_i$. If suffices to show $\sum a_i Z_i \sim N(0, \sum a_i^2)$. For $t \in \mathbb{R}$,

$$\mathbb{E} \exp\left(t \sum_{i=1}^n a_i Z_i\right) = \prod_{i=1}^n \mathbb{E} \exp(t a_i Z_i) = \prod_{i=1}^n \exp(t^2 a_i^2/2) = \exp\left(\frac{t^2}{2} \sum_{i=1}^n a_i\right)$$

since $Z_i \sim N(0,1)$. Hence

$$\sum_{i=1}^n a_i Z_i \sim N\left(0, \sum_{i=1}^n a_i^2\right).$$

Shifting back by m we get the distribution of Y.

For the complex case, write

$$Z_i = \frac{1}{\sqrt{2}}(U_i + iV_i) \quad \text{and} \quad a_i = \alpha_i + i\beta_i$$

with U_i, V_i independent $N(0, 10$. Then $\mathbb{E}Z_i = 1$ and $\mathbb{E}|Z_i|^2 = 1$. And we can prove that $\mathrm{Re}(a_i Z_i)$ and $\mathrm{Im}(a_i Z_i)$ are real linear combinations of the independent real Gaussians U_i, V_i. Hence $(\mathrm{Re} \sum b_i Z_i, \mathrm{Im} \sum a_i Z_i)$ is a real 2D Gaussian vector, so $\sum a_i Z_i$ is a complex Gaussian. □

Exercise 2.1.10. (Rotation invariance). Let $X = (X_1, \ldots, X_n)$ be an $\mathbb{R}^n$-valued random variable, where $X_1, \ldots, X_n \equiv N(0,1)_{\mathbf{R}}$ are iid real gaussians. Show that for any orthogonal matrix $U \in O(n)$, $UX \equiv X$.

Show that the same claim holds for complex gaussians (so X is now $\mathbb{C}^n$-valued), and with the orthogonal group $O(n)$ replaced by the unitary group $U(n)$.

Exercise 2.1.11. Show that Theorem 2.1.12 is equivalent to the inequality

$$\mathbb{P}(X \in A)\mathbb{P}(X \notin A_\lambda) \leq C \exp(-c\lambda^2)$$

holding for all $\lambda > 0$ and all measurable sets A, where $X = (X_1, \ldots, X_n)$ is an $\mathbb{R}^n$-valued random variable with iid gaussian components $X_1, \ldots, X_n \equiv N(0,1)_{\mathbb{R}}$, and A_λ is the λ-neighborhood of A.

2.2 The central limit theorem

2.2.1 Reductions.

Exercise 2.2.1. Show that Z_n does not converge in probability or in the almost sure sense. (Hint: the intuition here is that for two very different values $n_1 \ll n_2$ of n, the quantities Z_{n_1} and Z_{n_2} are almost independent of each other, since the bulk of the sum S_{n_2} is determined by those X_n with $n > n_1$. Now make this intuition precise.)

Exercise 2.2.2. Use Stirlings formula (Section 1.2) to verify the central limit theorem in the case when X is a Bernoulli distribution, taking the values 0 and 1 only. (This is a variant of Exercise 1.2.2 or Exercise 2.1.2. It is easy to see that once one does this, one can rescale and handle any other two-valued distribution also.)

2.2.2 The Fourier method.

Exercise 2.2.3. Use Exercise 2.1.9 to verify the central limit theorem in the case when X is gaussian.

Exercise 2.2.4. (Linearity of convergence) Let V be a finite-dimensional real or complex vector space, X_n, Y_n be sequences of V-valued random variables (not necessarily independent), and let X, Y be another pair of V-valued random variables. Let c_n, d_n be scalars converging to c, d respectively.

(i) If X_n converges in distribution to X, and Y_n converges in distribution to Y, and at least one of X, Y is deterministic, show that $c_n X_n + d_n Y_n$ converges in distribution to $cX + dY$.

(ii) If X_n converges in probability to X, and Y_n converges in probability to Y, show that $c_n X_n + d_n Y_n$ converges in probability to $cX + dY$.

(iii) If X_n converges almost surely to X, and Y_n converges almost surely Y, show that $c_n X_n + d_n Y_n$ converges almost surely to $cX + dY$.

Show that the first part of the exercise can fail if X, Y are not deterministic.

Exercise 2.2.5. Show that the normal distribution $N(\mu, \sigma^2)_{\mathbb{R}}$ has characteristic function $F(t) = e^{it\mu} e^{-\sigma^2 t^2/2}$.

Proof. Since the density function of $X \sim N(\mu, \sigma^2)$ is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right),$$

the characteristic function of X is

$$F(t) = \mathbb{E}e^{itX} = \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(itx - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)\, dx.$$

By some computation we have the exponent

$$itx - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 = -\frac{1}{2}\left(\frac{x-\mu}{\sigma} - it\right)^2 + it\mu - \frac{t^2}{2}.$$

So

$$F(t) = \frac{e^{it\mu} e^{-\sigma^2 t^2/2}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma} - it\right)^2\right)\, dx.$$

By substituting $y = \frac{x-\mu}{\sigma} - it$ and using the fact that $\int_{-\infty}^{\infty} \exp(-y^2/2)\, dy = \sqrt{2\pi}$, we have the desired result. $\square$

Exercise 2.2.6. (Riemann-Lebesgue lemma) Show that if X is an absolutely continuous random variable taking values in $\mathbb{R}^d$ or $\mathbb{C}^d$, then $F_X(t) \to 0$ as $t \to \infty$. Show that the claim can fail when the absolute continuity hypothesis is dropped.

Exercise 2.2.7. Show that the characteristic function F_X of a random variable X taking values in $\mathbb{R}^d$ or $\mathbb{C}^d$ is in fact uniformly continuous on its domain.

Exercise 2.2.8. (Taylor expansion of characteristic function) Let X be a real random variable with finite k^{th} moment for some $k \geq 1$. Show that F_X is k times continuously differentiable, and one has the partial Taylor expansion

$$F_X(t) = \sum_{j=0}^{k} \frac{(it)^j}{j!} \mathbb{E}X^j + o(|t|^k)$$

where $o(|t|^k)$ is a quantity that goes to zero as $t \to 0$, times $|t|^k$. In particular, we have

$$\frac{d^j}{dt^j} F_X(t) = i^j \mathbb{E}X^j$$

for all $0 \leq j \leq k$.

Proof. Consider the Taylor expansion of e^{itx} at $t = 0$, for fixed x, we have

$$e^{itx} = \sum_{j=0}^{k} \frac{(itx)^j}{j!} + r_k(t, x)$$

where the remainder

$$e_k(t, x) = \frac{f^{(k+1)}(y)}{(k+1)!} t^{k+1} \quad \text{for some } y \in (0, t).$$

Since $f^{(k+1)}(t) = (ix)^{k+1} e^{itx}$,

$$\frac{|r_k(t, x)|}{|t|^k} = \frac{|x|^{k+1}}{(k+1)!} |t| \to 0 \quad \text{as } t \to 0.$$

So

$$r_k(t, x) = o(|t|^k).$$

Let $x = X$ and take expectation, we have the desired form. □

Exercise 2.2.9. Establish (2.32) in the case that X is sub-Gaussian, and show that the series converges locally uniformly in t.